Funded by **NorrlandsNavet**

Living Big in a Tiny House. Design of Tiny Villages for the North

Final report

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Summary

Living Big in a Tiny House investigates how Compact Living Communities (CLCs), or tiny villages, can support sustainable and attractive housing development in northern Sweden. The project addressed three interconnected challenges: accommodating user groups with different housing needs, overcoming regul atory and infrastructure barriers, and designing compact dwellings suitable for Arctic conditions. Attention is given to the tension between lifestyle-oriented residents seeking alternative ways of living and temporary industry workers requiring affordable and practical accommodation.

A multi-phase design research methodology, combining Research-through-Design, participatory design, iterative co-creation, and scenario-based evaluation, helped structure four workshops involving municipalities, industry representatives, researchers, manufacturers, entrepreneurs, and potential users during 2025.

Workshop data were analyzed using Reflexive Thematic Analysis and Cross-Case Comparative Analysis. Recurring topics were coded and assessed through Trend Scores, Frequency Scores, Intensity Scores, and Evidence Counts, enabling comparison across workshops and stakeholder groups. Five persistent themes emerged: Spatial Design, Infrastructure, Social Dynamics, Governance and Regulation, and Mobility. Results reveal a recurring tension between user attractiveness and implementation feasibility, particularly concerning ownership models, financing mechanisms, infrastructure provision, accessibility, and regulatory compliance. Potential residents prioritized spatial quality, wellbeing, and community life, while institutional actors focused on governance, implementation, and technical feasibility.

The project delivers an actionable dataset and planning insights that support evidence-based development of compact living communities in Arctic regions.

Keywords: Tiny villages, compact living communities, tiny houses, sustainable communities, participatory design.

Introduction

Northern Sweden is undergoing significant industrial and demographic transformation driven by the expansion of green industries and the transition toward a more sustainable economy. These developments have increased the demand for housing solutions that can support population growth while responding to environmental challenges, resource constraints, and the particular conditions of Arctic regions. Within this context, Compact Living Communities (CLCs), or tiny villages[1], have emerged as a potential approach for providing housing that combines resource efficiency, adaptability, and quality of life.

The project Living Big in a Tiny House explores how design and architecture can contribute to sustainable living in northern regions. Grounded in participatory co-design, the project recognizes the importance of involving stakeholders in shaping future living environments, thereby strengthening both ownership and acceptance of proposed solutions [2] At the same time, it considers how compact housing can respond to demanding climatic conditions while maintaining comfort, functionality, and sustainability[3,4]

Among the groups attracted to compact living are downshifters, who value proximity to nature, reduced consumption, self-sufficiency, and alternative lifestyles [5]. However, the industrial expansion taking place across northern Sweden has also created demand for housing for temporary and medium-term workers. While the former may prioritize wellbeing, community, and environmental values, the latter often emphasize affordability, accessibility, and proximity to employment. Reconciling these differing expectations represents a central challenge for the development of CLCs.

Beyond user diversity, implementation is constrained by ownership models, regulations, infrastructure provision, mobility, and governance. Previous research highlights the importance of aligning sustainable solutions with users' everyday practices and values [6,7]. Consequently, this project approaches compact living communities as socio-technical systems in which spatial design, infrastructure, governance, mobility, and social interaction are closely interconnected.

The objective of the project is to identify opportunities, barriers, and implementation pathways for developing attractive, feasible, and sustainable compact living communities in Arctic regions through participatory design and collaborative knowledge creation.

Problem

The project addresses the problem of finding thresholds and barriers for adopting compact living for target populations with diverging needs. In this regard, there is a set of tensions between user profiles and the lack of awareness of the regulatory and technical barriers for tiny living in the arctic to happen.

The first tension level is a demographic one between downshifters and industry workers.

Downshifters are individuals seeking to live life at a slower pace driven by reduced consumption of goods and services; people whose ambitions allow them to adopt stationary-but-mobile lifestyle while pursuing creative and altruistic goals.

Industry workers are usually employed under short-term and medium-term contracts. This target population operates at a different pace and often follows different ambitions when compared to downshifters. They just need comfortable accommodation close to the workplace.

The gap appears when reconciling these dimensions, as planning and building CLCs in the arctic might require finding an average threshold to make them feasible in order to satisfy the needs of both target groups.

The second tension level lies in the regulatory field. Challenges appear when thinking about planning and building CLCs in places where basic infrastructure (energy, wastewater, tap water) is not present yet. The issue is a challenge for municipalities in terms of allocating resources for small communities, which in turn makes it unviable for said communities to afford the resulting cost of using those infrastructures because of the high cost it demands to build and maintain them.

The third tension level appears when thinking about compact homes for arctic weather. Under arctic winter conditions, tiny houses need to comply with minimum standards of insulation, thermal comfort and air quality, which might become challenging in terms of cost, not technology, mostly when thinking about passive or self-sufficient constructions.

The project addresses the above tensions by tackling the goals in the section below.

Goals

Living Big in a Tiny House was structured around following drivers:

Driver 1: Sustainable Nordic Tiny House Prototyping & Policy Innovation

To design and prototype material-efficient, energy-optimized tiny houses tailored for northern climates, while concurrently exploring circular economy recycling models and developing regulatory frameworks for mobile tiny house property building permits.

Driver 2: Co-Design & Collaborative Circular Networks for CLCs

To establish inclusive co-design processes with stakeholders for sustainable CLCs that foster social interaction, while simultaneously building a circular network of local entrepreneurs (SMEs), builders, and suppliers to share knowledge and materials.

Driver 3: Scientific contribution

To advance knowledge on how compact residential spaces can be designed to integrate renewable energy systems while optimizing functionality, aesthetics, and human well-being under arctic conditions.

These drivers and their connection to the tensions described in previous sections define the methodological framework adopted in the project, which focuses on participative co-creation as element for identifying trending opportunities, challenges and pathways for finding a compromise between living quality and feasibility of CLCs that satisfy manifold needs.

Methodology

The project used a Multi-Phase Methods Design based on sequential-iterative phases in which research-through-design[8,9], participatory design, and iterative prototyping cycles create an adaptive, phase-based design research framework.

With Iterative Co-Creation as pillar for finding challenges and opportunities[10], the project used Participatory Design as environment to implement co-creation and living-lab environments [11–13] which are believed to help gather manifold insights from diverse stakeholders including potential users, scholars, industry specialists and policymakers.

This strong stakeholder integration benefits from experiences from both the industry and policymakers explored jointly in participative workshops in which SMEs, community groups, and regulatory bodies share knowledge creation across research and practice. In such a scenario, stakeholders help shape and validate outcomes that can be defined as an action research approach.

The methodology is implemented in three stages. Stage one addresses the issue of assessing a set of desirable services and activities that can take place in a CLC and a set of speculative planning scenarios to discuss with stakeholders and the community in general through workshops.

Stage two establishes criteria for inviting participants and designing workshops that take into consideration a) hypothetical scenarios selected by the consortium [14] and b) enriched scenarios fed by cumulative input as workshops take place.

Stage three correlates the results of every workshop and performs a qualitative analysis of the contributions made by the participants. By contrasting the evolution of CLC scenarios and the input from the participants in the collaborative workshops, the consortium establishes a basis for defining design-to-implementation perspectives to potentially benefit, **a)** companies that produce compact living environments (attefalshus) and, **b)** bodies and associations interested in developing tiny living as a community living paradigm.

The project team produced digital scenarios using Devolv, Adobe Photoshop, and Rhinoceros that became interactive using visual collaboration platforms like Miro and Mural. Online sessions used Ms Teams and Zoom as environments for exchange.

The data gathered throughout the workshops uses reflexive thematic analysis (RTA) to code recurring concepts across transcripts, group them into themes, and interpret meaning and connections [15,16]. Cross-case comparative analysis (CCCA) is used to compare topics across workshop transcripts [17,18] and identify persistency and topic patterns that later become feeders for identifying trends and their relevance (Trend Score). Frequency scores and intensity scores across topics help measure how long and how deep conversations were on specific issues, which helps identify drivers for action.

The chosen methodology reflects the complexity and contextual specificity of the research challenge. By **c**ombining participatory design, iterative prototyping, and scenario-based

evaluation, the project ensures that its output is socially anchored and environmentally responsive while keeping feasibility in perspective for local stakeholders.

Implementation

The operationalization of our adaptive, phase-based design research framework produced four workshops. The sessions spanned the fall of 2025 starting with an online workshop in September, then two physical workshops took place in Luleå and Kiruna in October and November. The closing workshop took place online on December 5, 2025. The output of every workshop is described below.

Workshop 01, September 2025 (On-line).

The workshop was structured around three "CLC scenarios". While all scenarios were proposed using a plot 16 km north of Luleå, each scenario had its own variables for planning a CLC. These variables were set based on a) site topography, b) green masses (trees, bushes), c) proximity to roads and connectivity and, d) potential for developing off-grid scenarios, either partial or total, as illustrated in Fig. 1.



Fig. 1 A CLC scenario

The discussions during the workshop revolved around ownership (of tiny houses), legal structures, governance models, viability and feasibility, social composition, balance between private and community living, spatial layout and aesthetics, technical systems and infrastructure, regulations and planning approach.

37 seven people from municipalities and companies took part in the workshop.

Workshop 02, October 2025 (Luleå).

The second workshop used a participatory design approach fed from input from the first workshop to evolve into an iterative exercise in which participants composed the CLC using a set of components they combined in mockup. During the first workshop, clustering was seen as one of the necessary measures to make these CLCs more viable with the concept of tiny living

at the center of the discussion as a Lifestyle. As seen in Fig. 2, in this second workshop cluster design was at the center of the discussion.



Fig. 2 Clustering ideas from workshop 02

Five people -all potential users as downshifters- from various backgrounds ranging from economics and architecture to professionals in mining and material science were involved. The group was split in two, with one group focusing on cluster design and activity flows within clusters, while the second group focused on creating a cluster layout to plan the infrastructure connecting the clusters inside the CLC and the services/facilities in between clusters.

Workshop 03, November 2025 (Kiruna).

The third workshop gathered people from municipalities and the industry. The presence of Attefallshus manufactures was highly appreciated, as well as that of active actors in urban planning from Skellefteå and Kiruna. By exploring hypothetical arctic scenarios, as illustrated in Fig. 3, this workshop gathered experiences from the two previous workshops and turned them into discussion topics where – again – participatory design was the focus of the experience. 15 people participated in the workshop, which was split into two focus groups.

Each group had a moderator that would set up the topic, and participants had the opportunity to flexibly develop ideas. The experience yielded a series of patterns across the discussions that range from financing and bankability to mobility logistics and franchise logics.



Fig. 3 using parameters from previous workshops for participatory design.

Workshop 04, December 2025 (On-line).

The last workshop was held online from Luleå and used the input from the previous events. Workshop #4 brought a new scenario -set on a peninsula over the Lulea River near Kiruna- that proposes activities for leisure and wellbeing that potential users and stakeholders might find attractive.

Participants in this workshop used a collection of predefined CLC clusters they used for proposing planning alternatives benefiting from the features in the area (the river, a hill, the woods, an old barn). The discussions in this workshop were much deeper as they pay closer attention to aspects discussed in previous workshops, with feasibility, fairness, accessibility and attractiveness as core drivers for decision making and planning.

Workshop series analysis

Using RTA and CCCA, the research team compared topics across workshop transcripts and identify persistency and topic patterns to identify trends and their relevance as future drives to define design-to-implementation perspectives.

As result of implementing Cross-case comparative analysis (CCCA) across workshop transcripts (RTA) it was possible to identify topic trends and their relevance (Trend Score [*Tsc*]). Frequency scores [*Fsc*] and intensity scores [*Isc*] across topics helped to obtain indicators on topic conversation depth and length, which we think helps identify drivers for action. An evidence count indicator [*Ect*] will provide a measure of the number of times someone addressed the topic during workshop conversations.

Fig. 4 provides a *Tsc* comparison among workshops. A first analysis on *Tsc* shows a direct relation with *Ect* across workshops whose outcome is, in turn, driven by the attendants' profile to each workshop.

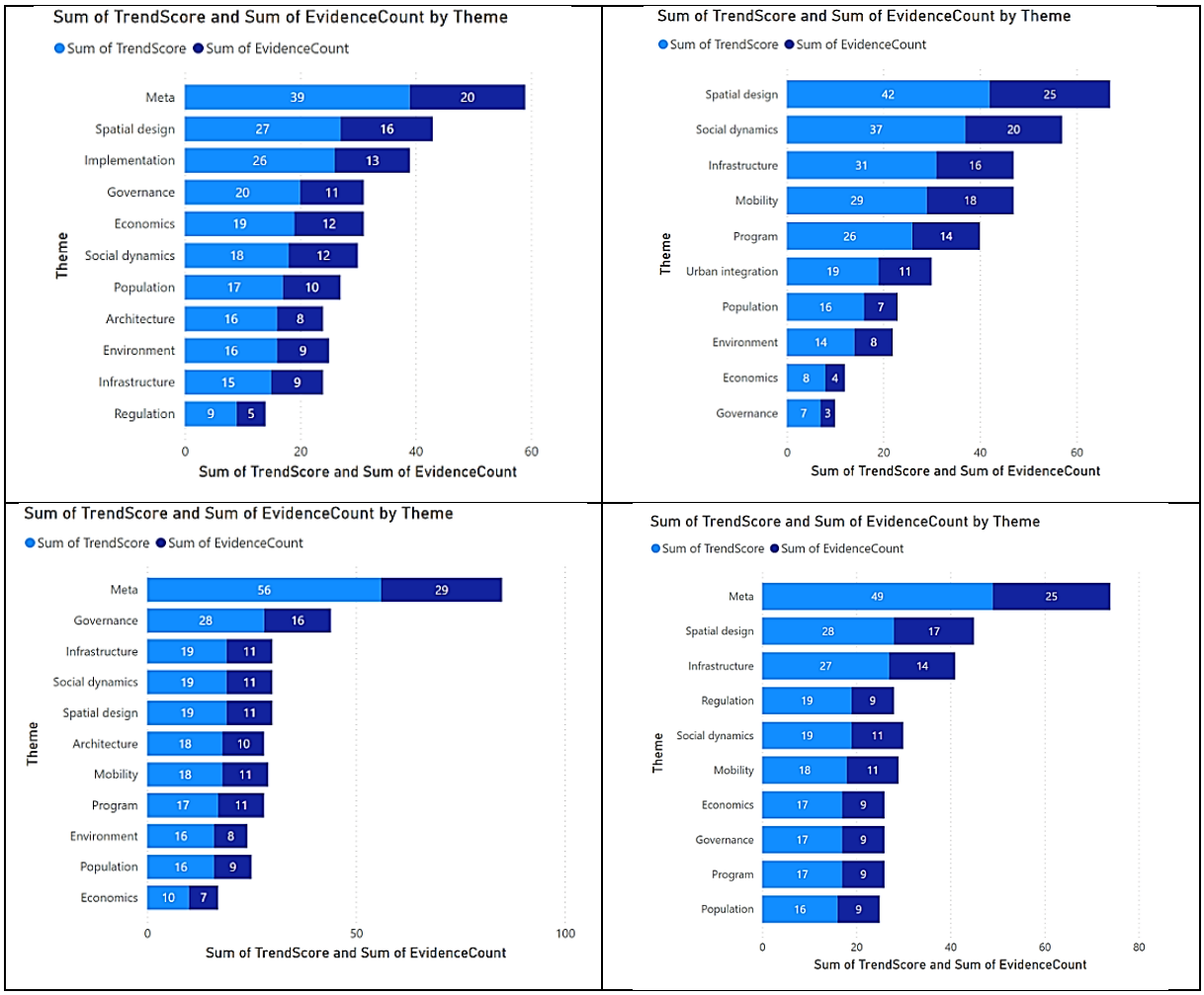


Fig. 4 Tsc comparison across workshops. Left-right, top to bottom. Workshops 1 to 4.

A thing to consider when analyzing data from simultaneous conversations is what has been found here as a Meta theme. A Meta theme embodies manifold discussions that randomly occur during conversations as participants tend to point out particularities that come as support for main themes. This is a common factor in WS 01, 03 & 04, where more people were involved.

In this regard, Meta subthemes appear as articulators rather than central discussions. The Meta theme disappears in WS 02 because of the small number of participants focusing on topics at the user level, rather than topics at feasibility levels, which was the case in the other three workshops. Fig. 5 illustrates the topics that made part of the Meta category across workshops.

Design vs feasibility is a persistent sub-theme within workshops along with *flexibilty vs infrastructure rigidity*, which finds a synonym as *community vs regulation*, in WS 01. This pattern reinforces the solidity of spatial design as trending theme throughout workshops 1,2 and 4. The break of this trend in WS 03 is caused by a bigger attendance of bankers, policy makers and manufacturers as compared to other workshops.

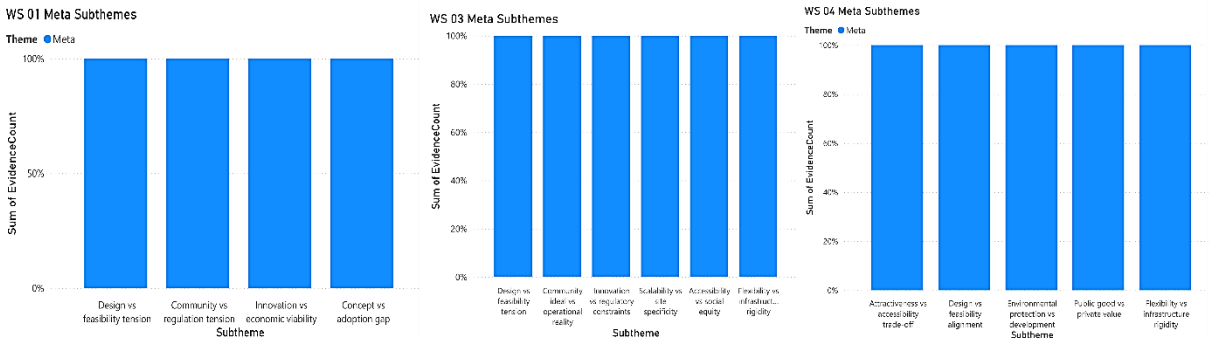


Fig. 5. Meta-related sub-themes in WS 01, 03, and 04.

Exchanges among participants to WS 03 focused more on governance and infrastructure because of the tension between design and feasibility whose achievability is important to parameterize aspects such as attractiveness, social dynamics or governance. Without *Spatial Design* input allowing to know what CLCs will look like or what they aim for, it might be difficult to set and align such parameters with issues like mobility, environmental impact, implementation, or infrastructure.

The role of *Spatial Design* throughout workshops creates another revelation. User perspective is key in aligning feasibility-related parameters (governance, economics, implementation, regulations…) with user needs to increase attractiveness. WS 02 invited people whose profile fits that of the end-user rather than that of the policy-maker, the manufacturer or the banker. People with this profile showed more interest in discussing spatial design.

A key element to observe is user perspective, which creates an interesting shift between WS 02 and the other workshops. Having only potential users as participants, WS 02 enriches participatory design with a relevant social component in which feasibility constraints are set aside to give way to what users consider important when considering moving in a CLC. During WS 02 participants' addressed *Spatial Design* from the perspective of subthemes like cluster configuration (the block), sun gains, and views. Social interaction and worker segregation came as subthemes of *Social Dynamics*, while concerns about *Infrastructure* focused on the availability of a seamless provision of energy and sanitation systems.

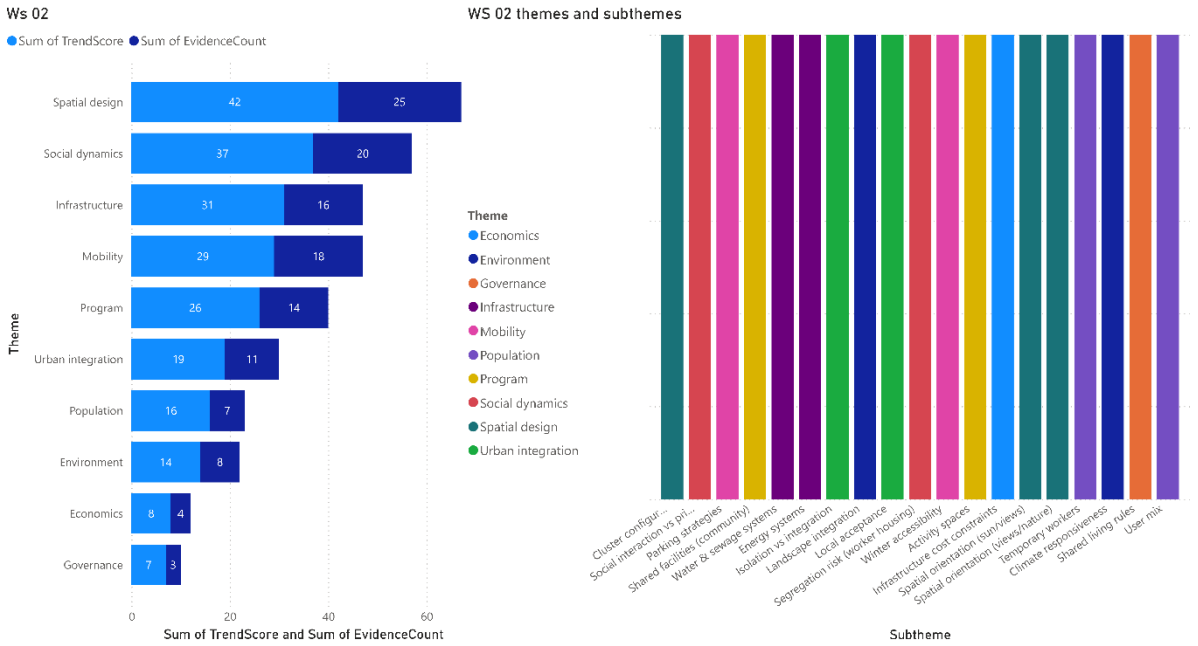


Fig. 6 WS 02, themes and subthemes

The output from WS 02 sets several *Spatial-Design*-driven priorities potential users consider important if they are to move in a CLC. As shown in Fig. 7, the intensity score [*Isc*] for the five top *Tsc* ranking themes besides *Spatial Design* shows that people participating in this workshop dedicated an important part of their time to discussing these topics while making and implementing design decisions, rather than dedicating time to discuss financial feasibility or regulatory frameworks.

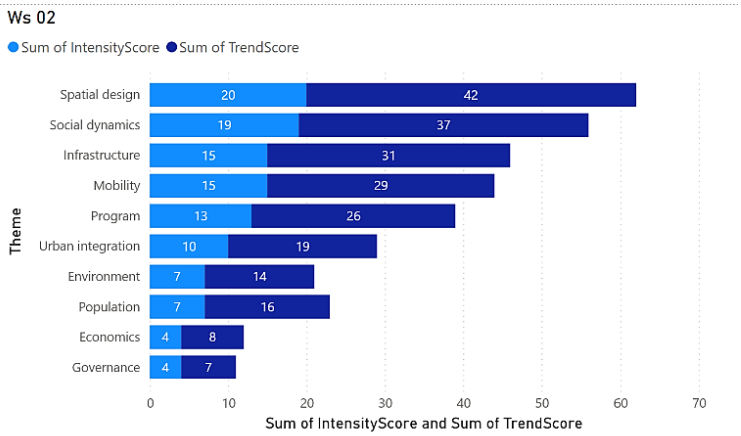


Fig. 7 WSC 02, Intensity Score [Isc] indicators

The output above reveals a dichotomy between interests at different participatory levels: discussions involving policy makers, bankers, scholars, and manufacturers tend to steer towards governance, infrastructure and implementation rather than to focus on user satisfaction as driving parameter for the former. The full dataset dashboard is available in Appendix 1 of this report.

Project Results

Engagement

One of the key challenges throughout the project was to attract the attention of key actors who could find a reciprocal benefit from participating in the workshops. Before the workshop series started a survey was conducted to identify potential participants from manifold backgrounds in an attempt to provide as much diversity and views as possible, as Fig. 8 illustrates

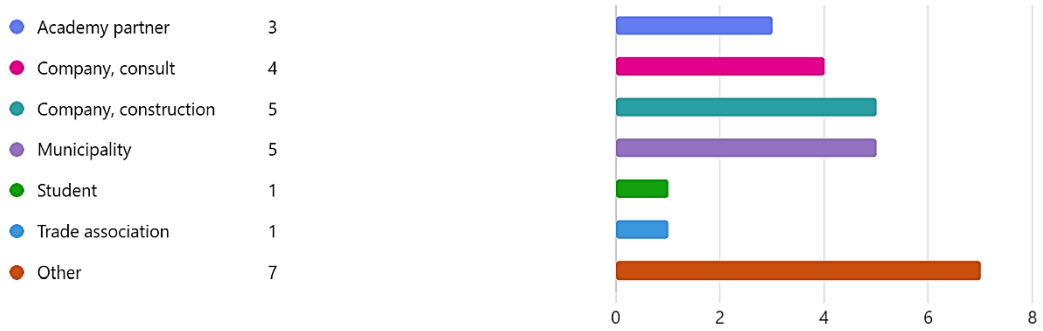


Fig. 8. Overall workshop expected attendance.

As workshops were launched, the consortium attempted to reach a wider audience through repeated invitations. The real attendance and the expected attendance from specific invitations are measured through an attendance realization indicator whose results are illustrated in Fig. 9 and are calculated as follows:

$$Ar = \frac{At}{26} x100$$

Where *Ar*= Attendance realization; *At* = Actual attendance.

These results revealed that geographic location and availability can be considered as the main barriers for stakeholder engagement in the workshops, which is especially sensible for people working both in industry and academia, where schedules are usually tight.

The participation of people from academia was, however, bigger than expected for WS 02, which in turn delivered more focused results when compared to workshops with a bigger attendance.

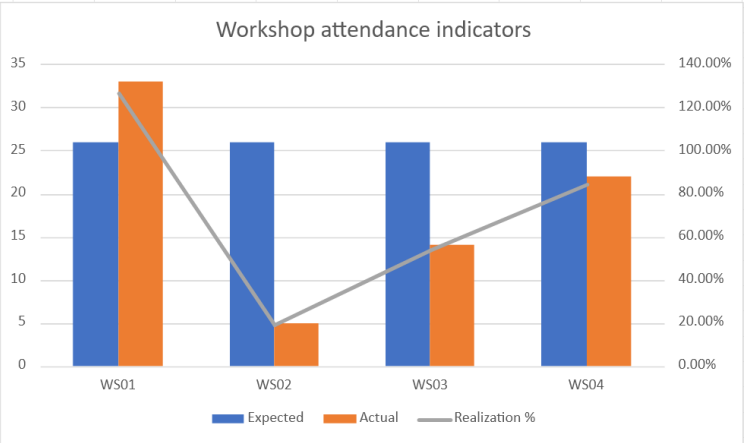


Fig. 9. Workshop attendance indicator

Output 1. Actionable data

As response to the first two tension levels in the project's problem statement, the workshop series produced an extensive dataset with information that can be later transformed into user-oriented CLC planning and implementation strategies.

By analyzing discussion across workshops using reflexive thematic analysis and cross-case comparative analysis, the workshop series delivers a dashboard featuring Trend Scores, Intensity Scores, Evidence Counts and Frequency Scores. These scores function as actionable parameters for better planning CLCs from both industry and societal perspectives.

The manifold scores revealed five trending topics planners should address when planning a compact living settlement (CLC) or tiny village.

Spatial Design encompasses a cluster of subthemes including CLC cluster configuration, orientation and interaction with nature in terms of thermal comfort, views and landscape. These subthemes are directly connected with the concept of wellbeing as driver for attractiveness.

Infrastructure. The resulting parameter of Spatial Design, infrastructure was at the center of several conversations with subthemes including infrastructure decentralization, infrastructure-driven village design, off-grid solutions, and soil adaptation.

Social dynamics, acquires a lot of importance as one of the key drivers in CLC attractiveness. The exchanges among participants underlined community behavior, private vs shared space social interaction, and segregation risks as key aspects to consider from a social impact point of view.

Governance shares its ranking with *Regulation* as key drivers for planning, with issues like ownership rights, construction permits, liability and risk allocation, and housing models (as business) being at the focus of conversations when participants addressed the topic

Mobility comes under the spot in conversation when the issue of getting to and from the CLC arises. Subthemes across conversation focused on strategies for winter accessibility, parking, road hierarches, and emergency vehicle access as elements to consider when discussing CLC attractivity.

A dashboard in Appendix 1 provides detailed information about themes and subthemes addressed throughout workshops and their relevance measured in terms of *Tsc* and *Isc.*

Output 2. Master's theses and student output

As a contribution to Driver 3 (Scientific Contribution), the project generated two master's thesis projects that expanded the investigation of challenges identified through the workshops and the project's problem statement. These studies provided complementary evidence on both the technical and social dimensions of compact living communities.

In *"Termisk prestanda i tiny houses: en jamf¨orande studie av olika ¨materialval",* Roxane Ndayambaje investigated how material choices and construction solutions affect the thermal performance of tiny houses. Her study concluded that energy simulations are a valuable tool for evaluating and optimizing tiny-house designs.

A second study by Albin Rieeem, titled *"Tiny Houses i Sverige: En kvantitativ studie om acceptabel enkelhet,boendekomfort och social acceptans"*, investigated public perceptions of tiny houses in Sweden, focusing on acceptable simplicity, living comfort, and social acceptance. The study concluded that tiny houses have potential in Sweden but require improved residential comfort and broader cultural acceptance to achieve wider adoption.

Output 3. Follow-up project grants and articles

The results from the workshops solidified concepts and ideas within the consortium, which helped bring together manifold actors whose competence and insight provided tools for assembling a follow up project that received funds from Vinnova.

This new project is targeting users in Norbotten's production apparatus as inhabitant influx connected to industrial operations and resulting side services (healthcare, education, lodging) are expected to grow over the next decades.

As part of Driver 3, the findings documented in this report are also being developed into a peer-reviewed journal article. The article expands on the methodological framework combining participatory design, Reflexive Thematic Analysis, and Cross-Case Comparative Analysis, and discusses how stakeholder-derived evidence can support the planning and implementation of Compact Living Communities in Arctic regions.

Conclusion

The project set out to explore the opportunities and barriers associated with developing Compact Living Communities (CLCs) in northern Sweden. Three challenges guided the work: accommodating the needs of diverse user groups, understanding regulatory and infrastructure barriers, and identifying requirements for compact housing under Arctic conditions.

The workshop series demonstrated that the feasibility of CLCs depends on balancing user attractiveness with practical implementation. While potential residents consistently emphasized spatial quality, access to nature, privacy, community life, and wellbeing, institutional actors focused on infrastructure, governance, financing, and regulatory frameworks. Rather than representing opposing views, these perspectives were found to be complementary and necessary for the successful implementation of compact living environments.

The project achieved its objectives by establishing a collaborative platform where municipalities, researchers, industry representatives, manufacturers, entrepreneurs, and potential users could jointly discuss and evaluate future CLC scenarios. Through this process, a substantial qualitative dataset was generated and analyzed. The analysis identified five recurring themes that consistently influenced planning decisions: Spatial Design, Infrastructure, Social Dynamics, Governance and Regulation, and Mobility. These themes provide a practical framework for future planning and development initiatives.

The results indicate that compact living communities should not be approached solely as housing products. Their success depends on integrating settlement design, infrastructure, governance, mobility, and social interaction into a coherent planning strategy. In this regard, spatial design emerged as a key enabling factor capable of aligning user expectations with implementation requirements.

Beyond the workshop outcomes, the project delivered an actionable dataset, strengthened collaboration among regional stakeholders, and contributed knowledge that has already supported the development of follow-up initiatives. The findings provide a foundation for future research, policy development, and implementation efforts aimed at creating attractive, sustainable, and feasible compact living communities in Arctic regions.

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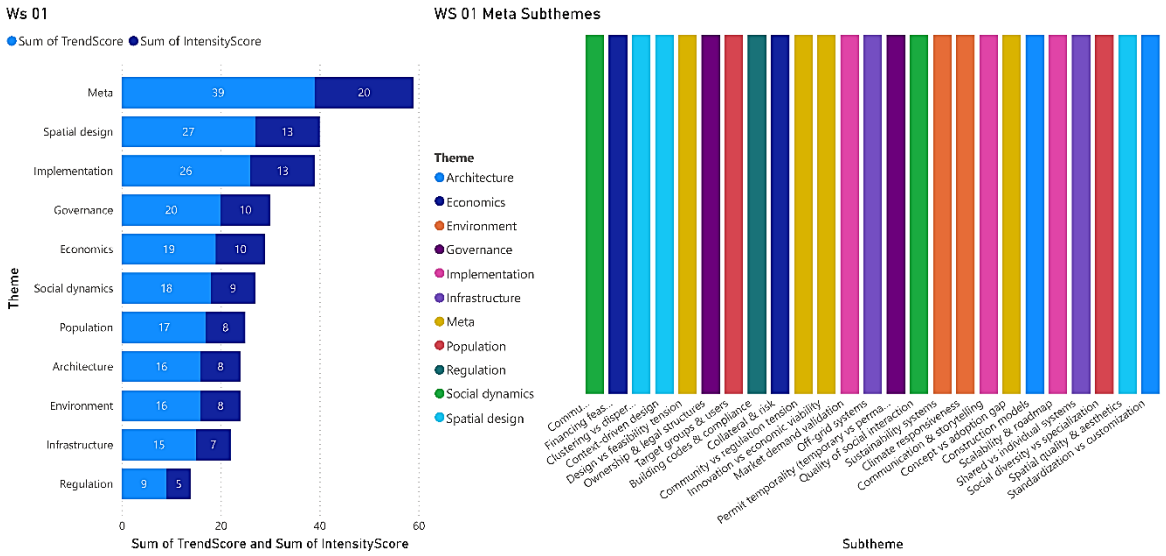
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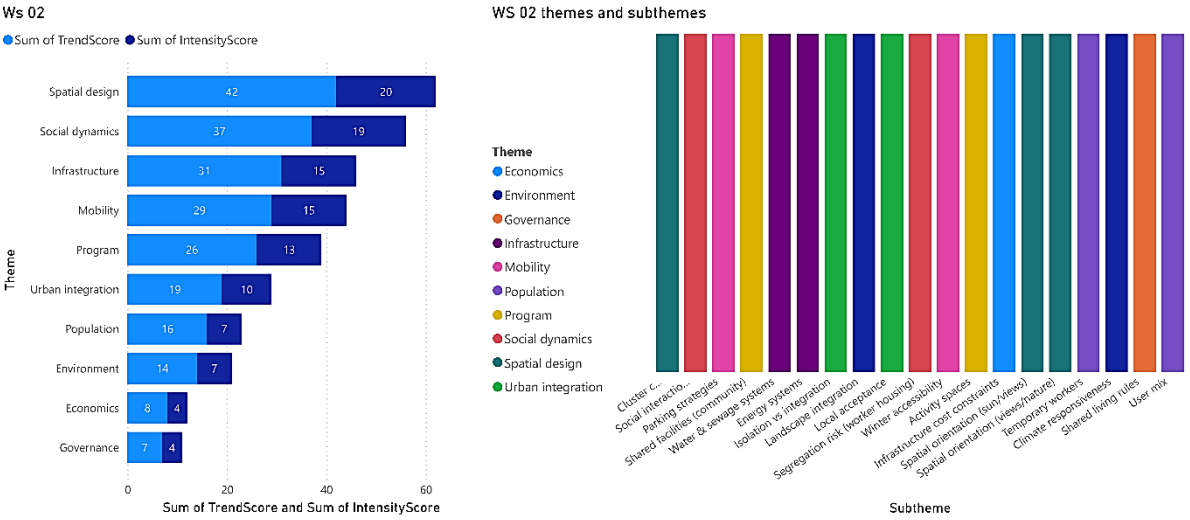
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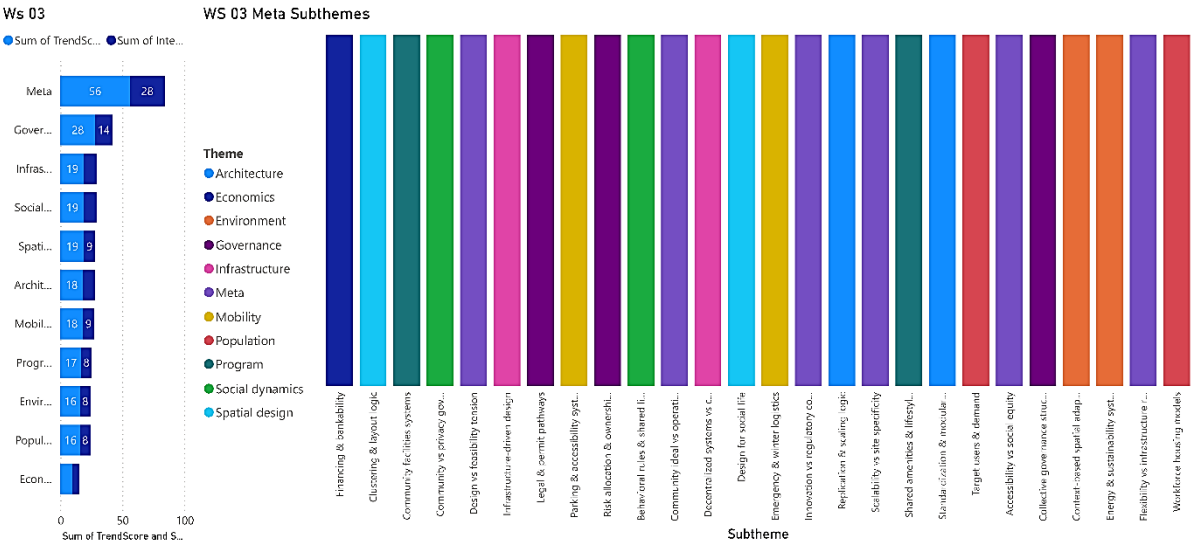
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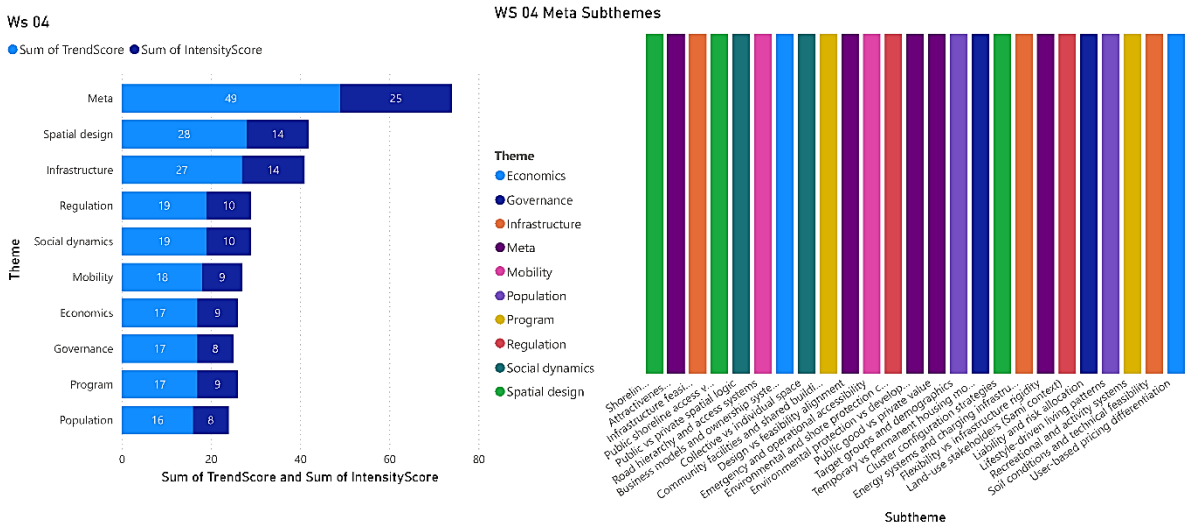
Appendix 1

Workshop dataset dashboard






Ws 03
Sum of TrendSc...
Sum of Inte...
Meta
Gover...
Infras...
Social...
Spati...
Archit...
Mobil...
Progr...
Envir...
Popul...
Econ...
56
28
28
14
19
19
19
9
18
18
9
17
8
16
8
16
8
0
50
100
Sum of TrendScore and S...
WS 03 Meta Subthemes
Theme
Architecture
Economics
Environment
Governance
Infrastructure
Meta
Mobility
Population
Program
Social dynamics
Spatial design
Financing & bankability
Clustering & layout logic
Community facilities systems
Community vs privacy gov...
Design vs feasibility tension
Infrastructure-driven design
Legal & permit pathways
Parking & accessibility syst...
Risk allocation & ownershi...
Behavioral rules & shared li...
Community deal vs operati...
Decentralized systems vs c...
Design for social life
Emergency & winter logistics
Innovation vs regulatory co...
Replication & scaling logic
Scalability vs site specificity
Shared amenities & lifestyl...
Standardization & modular ...
Target users & demand
Accessibility vs socio equity
Collective governance struc...
Context-based spatial adap...
Energy & sustainability syst...
Flexibility vs infrastructure ...
Workforce housing models
Subtheme


Ws 04
Sum of TrendScore
Sum of IntensityScore
Meta
Spatial design
Infrastructure
Regulation
Social dynamics
Mobility
Economics
Governance
Program
Population
49
25
28
14
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10
18
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17
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16
8
Theme
0
20
40
60
80
Sum of TrendScore and Sum of IntensityScore
WS 04 Meta Subthemes
Theme
Economics
Governance
Infrastructure
Meta
Mobility
Population
Program
Regulation
Social dynamics
Spatial design
Shorelin...
Attractiveness...
Infrastructure feasi...
Public shoreline access v...
Public vs private spatial logic
Road hierarchy and access systems
Business models and ownership syste...
Collective vs individual space
Community facilities and shared build...
Design vs feasibility alignment
Emergency and operational accessibility
Environmental and shore protection c...
Environmental protection vs develop...
Public goods vs private value
Target groups and demographics
Temporary vs permanent housing mo...
Cluster configuration strategies
Energy systems and charging infrastru...
Flexibility vs infrastructure rigidity
Land-use stakeholders (Sami context)
Liability and risk allocation
Lifestyle-driven living patterns
Recreational and activity systems
Soil conditions and technical feasibility
User-based pricing differentiation
Subtheme